Asutosh Mishra

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Summary

Final year BS-MS (Research) Physics student with a minor in Computational Data Science at IISER Berhampur, experienced in **LLMs, NLP, and production backend systems**. Skilled in **Python, PyTorch, FastAPI, and RAG-based AI pipelines**, with hands-on experience building scalable, real-world AI products. Qualified **GATE Data Science & AI 2026 (AIR 1377)**. Seeking **AI Research / Backend roles**.

Technical Skills

Languages: Python, C/C++, SQL

AI & GenAI: Large Language Models (LLMs), LangChain, RAG, Agentic AI, PyTorch, TensorFlow, Hugging Face

Data Science and Visualization: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, SciPy, Streamlit, Gradio

Backend & Cloud: FastAPI, MongoDB, Docker, Grafana, Git

Mathematics: Linear Algebra, Probability, Statistics, Numerical Simulation

Experience and Projects

Data Science Intern February 2025 – January 2026

ATTOOM AI TECHNOLOGY PRIVATE LIMITED RESTful API development, Agentic AI, FastAPI, Grafana

- **Sole backend developer** for SageCV (www.sagecv.com), an AI-powered career guidance platform; architected and built the entire backend from scratch, delivering **production-grade, scalable APIs**.
- Designed modular **FastAPI** endpoints powering **AI agents** for skill extraction, scoring, course recommendation, and JD-aligned resume keyword optimization; integrated retrieval-augmented generation **(RAG)** with **Serper-based web search** for automated **project plagiarism detection.**
- Optimized APIs with concurrency (FastAPI async calls) for improved throughput.
- Implemented an automated **Meta (Instagram) content pipeline** leveraging LLM-based planning for end-to-end post scheduling and publishing.
- Integrated real-time system monitoring, request logging, and usage metrics via **Grafana**.

Summer Intern May 2024 – July 2024

LaRGo - LLM reseach group IISER Kolkata Python, LLMs, Prompt Engineering, NLP

- Conducted research under Dr. Kripabandhu Ghosh (IISER Kolkata) on evaluating the faithfulness of LLM explanations via **reproducibility analysis in high-stakes domains.** (Legal, Health and Finance).
- Developed a novel method termed ”ReQuesting” to benchmark the reproducibility of multiple SOTA LLMs like Gemini, Mixtral and Llama3-70b within these critical domains, primarily through prompt engineering techniques.
- The ”ReQuesting” method also enables models to identify suitable algorithms for task execution, thereby enhancing the trustworthiness and transparency of LLM outputs, which is particularly crucial in high-risk applications.
- **Preprint:** https://arxiv.org/pdf/2412.07781

Conference track project (Rank 3 at FIRE 2023) May 2023 – July 2023

Forum for Information Retrieval and Evaluation 2023 Python, Hugging Face Transformers

- Developed a **multi-label classifier** leveraging **BERT** architecture for the Artificial Intelligence on Social Media (AISoMe) track addressing public concerns towards vaccines at the FIRE 2023 conference.
- **Fine-tuned** the Covid-Twitter-BERT-v2 model on a subset of the CAVES dataset, provided by the conference, specifically customized for multi-label classification of vaccine tweets into concern categories involving 12 labels, where each tweet could possess multiple labels.
- Secured **3rd rank** in the AISoMe track with a macro F1 score of 0.70, performing within a 1% margin of the winning model (0.71).
- **Paper:** https://ceur-ws.org/Vol-3681/T1-11.pdf (Accepted at FIRE 2023)

Education

Indian Institute of Science Education and Research (IISER), Berhampur Berhampur, Odisha

BS-MS Physics major, Computational Data Science minor 2021 – 2026

Current Cumulative Performance Index (CPI): **8.4/10.0**

Ravenshaw HS School, Cuttack Cuttack, Odisha

+2, Science , PCMB 2018 – 2020